Sheet B

Q1: Solving by separation of variables and verifying solution

a) $\frac{dy}{dt} = 4t^3 \Rightarrow \int dy = \int 4t^3 dt$

$y(t) = t^4 + c, \quad y(0) = 1$

$1 = 0^4 + c \Rightarrow c = 1$

$y(t) = t^4 + 1$ Check $\frac{dy(t)}{dt} = 4t^3$ ✓

b) $(t+1)\frac{dy}{dt} = y \Rightarrow \int \frac{dy}{y} = \int \frac{dt}{t+1}$

$\Rightarrow \ln(y) = \ln(t+1) + c$

$y(t) = c(t+1), \quad y(1) = 4$

$4 = c(1+1) = 2c$

$c = 2$

$\therefore y = 2(t+1)$

Check: solution:

$\frac{dy}{dt} = \frac{d}{dt} 2(t+1) \Rightarrow \frac{dy}{dt} = 2$

Multiply RHS by $\frac{y}{y} = \frac{y}{2(t+1)}$

$$\frac{dy}{dt} = \frac{\cancel{2}y}{\cancel{2}(t+1)} \Rightarrow \frac{dy}{dt}(t+1) = y$$

(Or use implicit differentiation)

c) $$\frac{dy}{dt} = -\frac{2y}{(1+t)^3} \Rightarrow \int \frac{dy}{y} = -2\int \frac{dt}{(1+t)^3}$$

LHS: $\int \frac{dy}{y} = \ln(y) + c$

RHS: Let $1+t = u \Rightarrow \frac{du}{dt} = 1 \Rightarrow du = dt$

$$-2\int \frac{du}{u^3} = -2\int u^{-3}\,du = u^{-2} + c = \frac{1}{(1+t)^2} + c$$

$$\therefore \ln y = \frac{1}{(1+t)^2} + c \quad , \quad y(0) = 20$$

$$\ln(20) = 1 + c \Rightarrow c = \ln(20) - 1$$

$$\ln(y/20) = \frac{1}{(1+t)^2} - 1$$

$$y = 20 \cdot \exp\left(\frac{1}{(1+t)^2} - 1\right) = \frac{20}{e}\exp\left((1+t)^{-2}\right)$$

Use chain rule:

$$\frac{dy}{dt} = \underbrace{\frac{2c}{c} \cdot -2(1+t)^{-3} \cdot \exp((1+t)^{-2})}_{=y}$$

$$\therefore \quad \frac{dy}{dt} = \frac{-2y}{(1+t)^{-3}} \quad \checkmark$$

d) $e^t \frac{dy}{dt} = \sqrt{y} = y^{1/2} \implies \int y^{-1/2} dy = \int e^{-t} dt$

$\implies 2y^{1/2} = -e^{-t} + c, \quad y(0) = 4$

$\therefore \quad 2 \cdot 4^{1/2} = -e^{-0} + c$

$\implies 4 = -1 + c \implies c = 5$

$$y = \left(\frac{-e^{-t} + 5}{2}\right)^2 = \frac{(-e^{-t}+5)^2}{4} = \frac{(-e^{-t}+5)(-e^{-t}+5)}{4}$$

$$= \frac{e^{-2t} - 10e^{-t} + 25}{4}$$

$$= \frac{e^{-2t}}{4} - \frac{5}{2}e^{-t} + \frac{25}{4}$$

$$\frac{dy}{dt} = -\frac{1}{2}e^{-2t} + \frac{5}{2}e^{-t}$$

$=$

Mutiply both sides by e^t

$$e^t \frac{dy}{dt} = \underbrace{-\frac{1}{2}e^{-t} + \frac{5}{2}}_{\sqrt{y}}$$

$$= \frac{1}{2}e^{-t} + \frac{5}{2}$$

Q2: Partial fractions

a) $$\frac{1}{(x-2)(x+3)} = \frac{A}{(x-2)} + \frac{B}{(x+3)}$$

Multiply both sides by $(x-2)(x+3)$

$$1 = A(x+3) + B(x-2)$$
$$= Ax + 3A + Bx - 2B$$
$$= x(A+B) + 3A - 2B$$

$\therefore$ $A + B = 0 \Rightarrow A = -B$

$3A - 2B = 1 \Rightarrow 3A + 2A = 1 \Rightarrow 5A = 1 \quad A = \frac{1}{5}, \quad B = -\frac{1}{5}$

$$\Rightarrow \frac{1/5}{(x-2)} - \frac{1/5}{(x+3)}$$

$$\Rightarrow \frac{1}{5}\int \frac{1}{x-2}\,dx - \frac{1}{5}\int \frac{1}{x+3}\,dx = \frac{1}{5}\left(\ln(x-2) - \ln(x-3)\right) + C$$

b) $$\frac{1}{x^3 - 5x^2 + 6x} = \frac{1}{x(x^2 - 5x + 6)} = \frac{1}{x(x-2)(x-3)}$$

$$= \frac{A}{x} + \frac{B}{x-2} + \frac{C}{x-3}$$

Multiply both sides by $x(x-2)(x-3)$

$$1 = A(x-2)(x-3) + Bx(x-3) + Cx(x-2)$$

$$= Ax^2 - 5Ax + 6A + Bx^2 - 3Bx + Cx^2 - 2Cx$$

Collect x^2, x^1, x^0

$$1 = x^2(A+B+C) + x(-5A-3B-2C) + x^0(6A)$$

$$\therefore \begin{aligned} A+B+C &= 0 \\ -5A-3B-2C &= 0 \\ 6A &= 1 \end{aligned}$$

$$\therefore \quad A = 1/6, \quad B = -\frac{1}{2}, \quad C = \frac{1}{3}$$

$$\therefore \int \frac{1}{6x}\,dx + \int \frac{-1}{2(x+2)}\,dx + \int \frac{1}{3(x+3)}\,dx$$

$$= \frac{1}{6}\ln(x) - \frac{1}{2}\ln(x+2) + \frac{1}{3}\ln(x+3) + c$$

c) $$\frac{3x+11}{x^2-x-6} = \underbrace{\frac{3x+11}{(x+2)(x-3)} = \frac{A}{(x+2)} + \frac{B}{(x-3)}}$$

multiply both sides by $(x+2)(x-3)$

$$3x+11 = A(x-3) + B(x+2)$$

$$= Ax - 3A + Bx + 2B$$

$$= x(A+B) + x^0(2B-3A)$$

$\therefore \quad 3 = A + B \Rightarrow B = 3 - A$

$11 = 2B - 3A$

$\Rightarrow 11 = 6 - 2A - 3A = 6 - 5A$

$5 = -5A, \quad A = -1$

$\therefore B = 4$

$$\therefore \int \frac{-1}{x+2}\,dx + \int \frac{4}{x-3}\,dx = -\ln(x+2) + 4\ln(x-3) + C$$

Q3: Logistic model: $\frac{dN}{dt} = \beta N\left(1 - \frac{N}{K}\right)$

a) Model includes a density dependence which is linear in population size: $\frac{1}{N}\frac{dN}{dt} = \beta\left(1 - \frac{N}{K}\right)$

b) To solve, rewrite as: $\frac{dN}{dt} = \beta N\left(\frac{K-N}{K}\right)$

$$\Rightarrow \underbrace{\frac{dN}{N\left(\frac{K-N}{K}\right)}}_{} = \beta\,dt$$

Partial fractions needed!

$$\frac{1}{N\left(\frac{K-N}{K}\right)} = \frac{A}{N} + \frac{B}{\frac{K-N}{K}}$$

$$\Rightarrow 1 = A\left(\frac{k-N}{k}\right) + BN$$

$$= A - A\frac{N}{k} + BN$$

$$= N\left(\frac{-A}{k} + B\right) - A$$

$$\therefore A = 1$$

$$\therefore 1 = \frac{-1}{k} + B \Rightarrow B = \frac{1}{k}$$

$$\int \frac{1}{N} + \frac{1}{k-N}\, dN = \beta \int dt$$

$$\Rightarrow \ln(N) - \ln(k-N) = \beta t + C$$

$$= \ln(N/k-N) = \beta t + C$$

@ $t=0$ $\quad N = N_0$

$$\therefore \ln\left(\frac{N_0}{k-N_0}\right) = C$$

$$\frac{N}{k-N} = e^{\beta t} \cdot \frac{N_0}{k-N_0} \qquad \text{Let } J = \frac{k-N_0}{N_0}$$

Take 1/everything

$$\frac{k-N}{N} = \frac{J}{e^{\beta t}}$$

$$\Rightarrow \frac{k}{N} - 1 = \frac{J}{e^{\beta t}}$$

$$\Rightarrow N = \frac{k}{1 + Je^{-\beta t}}$$

Verify the solution, Let $u = 1 + Je^{-\beta t}$

$$\frac{dN}{dt} = \frac{dN}{du} \times \frac{-\beta J}{e^{\beta t}}$$

$$\frac{du}{dt} = -\beta J e^{-\beta t}$$

$$\therefore dt = \frac{du}{-\beta J} e^{\beta t}$$

$$= \frac{-\beta J}{e^{\beta t}} \frac{d}{du} \frac{k}{u}$$

$$= \frac{-\beta J}{e^{\beta t}} \cdot \frac{-k}{u^2} = \frac{k\beta J e^{-\beta t}}{(1 + Je^{-\beta t})^2}$$

Recall $N = \dfrac{k}{1 + Je^{-\beta t}}$ $\therefore$ $1 + Je^{-\beta t} = k/N$

and

$$\underbrace{Je^{-\beta t} = k/N - 1}_{\text{Substitute}}$$

$$\frac{dN}{dt} = \frac{k\beta(k/N - 1)}{(k/N)^2} = \frac{k^2\beta/N - k\beta}{k^2/N^2}$$

$$= \frac{k^2\beta/N}{k^2/N^2} - \frac{k\beta}{k^2/N^2} = \overbrace{\beta N - \frac{\beta N^2}{k}}^{\text{Benalli}}$$

$$= \underbrace{\beta N\left(1 - \frac{N}{k}\right)}_{\text{The original logistic model}}$$

Q4 : Bernoulli

Solve $\frac{dN}{dt} = \alpha N - \gamma N^2$

Let $Y = \frac{1}{N} \Rightarrow N = \frac{1}{Y}$

$$\frac{d}{dt}\frac{1}{Y} = \frac{\alpha}{Y} - \frac{\gamma}{Y^2}$$

Chain rule for LHS

$$\frac{-1}{Y^2}\frac{dY}{dt} = \frac{\alpha}{Y} - \frac{\gamma}{Y^2}$$

$$\frac{dY}{dt} = -\alpha Y + \gamma$$

$$\int \frac{dY}{-\alpha Y + \gamma} = \int dt$$

$$\Rightarrow \frac{-\ln(-\alpha Y + \gamma)}{\alpha} = t + c$$

@ $t = 0$, $N = 1$, $\therefore Y(0) = 1$ too

$$-\frac{1}{\alpha}\ln(-\alpha+\gamma) = c$$

$$\therefore \quad -\frac{1}{\alpha}\ln(-\alpha y+\gamma) + \frac{1}{\alpha}\ln(-\alpha+\gamma) = t$$

$$= \frac{1}{\alpha}\left(\ln(\gamma-\alpha) - \ln(\gamma-\alpha y)\right)$$

$$= \frac{1}{\alpha}\ln\left(\frac{\gamma-\alpha}{\gamma-y\alpha}\right) = t$$

$$\Rightarrow \frac{\gamma-\alpha}{\gamma-y\alpha} = e^{\alpha t} \Rightarrow \frac{\gamma-\alpha}{e^{\alpha t}} = \gamma - y\alpha$$

$$\Rightarrow \quad y = \frac{\gamma - \frac{\gamma-\alpha}{e^{\alpha t}}}{\alpha} \Rightarrow N = \frac{\alpha}{\gamma - \frac{\gamma-\alpha}{e^{\alpha t}}}$$

Multiply by $\frac{e^{\alpha t}}{e^{\alpha t}}$

$$\boxed{N = \frac{\alpha e^{\alpha t}}{\gamma e^{\alpha t} - \gamma + \alpha}}$$

For $N(0) = 1$, $J = k-1$

Original eqn $\frac{dN}{dt} = \alpha N - \gamma N^2$

$$= N(\alpha - \gamma N)$$

$$= \frac{N}{\alpha}\left(1 - \frac{\gamma N}{\alpha}\right)$$

$$\therefore \boxed{\beta = \frac{1}{\alpha}, \ k = \frac{\gamma}{\alpha}}$$

Substitute and verify.

Q5: logistic growth extension

$$\frac{dI}{dt} = \overbrace{(\alpha + \beta I)}^{\text{Rate of infection}} \underbrace{(N - I)}_{\text{Number of healthy trees}}$$

Rate of infection $= \alpha + \beta I$

$\therefore$ α = constant infection rate

βI = density dependent infection rate

Number of healthy trees $= N - I$

N = total trees

I = infected trees

Solve by separation of variables and partial fractions.

$$\frac{1}{(\alpha + \beta I)(N - I)} = \frac{A}{\alpha + \beta I} + \frac{B}{N - I}$$

$$1 = A(N - I) + B(\alpha + \beta I)$$

$$= AN - AI + B\alpha + B\beta I$$

$$= I(B\beta - A) + I^0(AN + B\alpha)$$

$\therefore$ $0 = B\beta - A$, $1 = AN + B\alpha$

$A = B\beta$ $\quad$ $1 = B\beta N + B\alpha = B(\beta N + \alpha)$

$$B = \frac{1}{\beta N + \alpha} \qquad A = \frac{\beta}{\beta N + \alpha}$$

$$\frac{\beta}{\beta N + \alpha} \int \frac{1}{\alpha + \beta I} dI + \frac{1}{\beta N + \alpha} \int \frac{1}{N - I} dI = \int dt$$

$$= \frac{\ln(\alpha + \beta I)}{\beta N + \alpha} + \frac{-\ln(N - I)}{\beta N + \alpha} = t + C$$

$$I(0) = 0$$

$$\frac{\ln(\alpha) - \ln(N)}{\beta N + \alpha} = C$$

$$\therefore \ln\left(\frac{\alpha + \beta I}{N - I}\right) = \overbrace{t(\beta N + \alpha)}^{F} + \ln \alpha - \ln N$$

$$\ln\left(\frac{(\alpha + \beta I) N}{(N - I)\alpha}\right) = F$$

$$\frac{N(\alpha + \beta I)}{\alpha(N - I)} = e^F \Rightarrow \frac{\alpha + \beta I}{N - I} = \frac{\alpha}{N} e^F$$

$$\Rightarrow I = \frac{\alpha N (e^F - 1)}{\beta N + \alpha e^F}$$

$$\boxed{= \frac{\alpha N (1 - \exp(-(\beta N + \alpha)t))}{\beta N \exp(-(\beta(N + \alpha)t) + \alpha}}$$

Wow